Sequential Testing for A/B Experiments

A Practical Guide in Three Acts

The problem. Every company running A/B tests faces a temptation: peek at the results before the experiment is over. Traditional statistics breaks when you do this — the false positive rate inflates far beyond the promised 5%.

The solution. *Sequential testing* provides statistical methods that remain valid no matter when or how often you check. Modern platforms like Eppo build this into their pipeline automatically.

This guide. Three acts, each with a simulation, intuitive explanations, and key formulas with plain-language translations:

1. **The Peeking Problem** — Why checking early is dangerous.
2. **The Eppo Solution** — How a modern platform solves it, including the recommended **hybrid approach**.
3. **DIY Alternatives** — What to do if you don't have Eppo, including how to implement the hybrid approach yourself.

Target audience. People comfortable with algebra and basic probability. No calculus required. Every formula is accompanied by a plain-language translation.

Act 1 — The Peeking Problem

The Motivation

Intuitive Explanation

You work at a company running an A/B test. Half your users see the old website, half see a new design. Every day, you check the dashboard: “Is the new design better?” After a

week, the dashboard shows $p = 0.03$ — that looks significant! You call it: the new design wins.

But here is the dirty secret of traditional statistics: **the more often you check, the more likely you are to see a false positive.** A test designed to have a 5% false positive rate can easily reach 20–30% if you peek at it repeatedly. One in four or five "winning" experiments might actually be noise.

This is not a theoretical concern. It is the single most common statistical mistake in online experimentation.

Simulation

What is being simulated: 1,000 A/B experiments where there is *no real effect* — any apparent difference is pure noise.

Two groups of observations (control and treatment) are generated from *identical* distributions. A standard t-test is computed after every new observation arrives.

A counter shows: "Number of times $p < 0.05$ so far." The user watches the p-value bounce around. Every time it dips below 0.05, it lights up red — a false positive.

Peeking Simulation

Run 500 A/A experiments (no real effect) and count how often you get a false positive under different peeking schedules.

Run Simulation

Why Does This Happen?

Intuitive Explanation

Traditional statistical tests promise: "If there is no real effect, I will falsely cry wolf at most 5% of the time." But this promise comes with fine print: **you may only look at the result once, at a pre-determined time.**

Every time you peek, you get another chance for randomness to fool you. It is like rolling a die repeatedly — the more rolls, the more likely you are to get a six eventually, even though any single roll has only a $1/6$ chance.

How Bad Is It?

Checking schedule	Actual false positive rate
Once at the end (as designed)	$\sim 5\%$
Daily for 1 week	$\sim 13\%$
Daily for 2 weeks	$\sim 19\%$
Daily for 4 weeks	$\sim 25\%$
Continuously (every observation)	Can exceed 30%

If you check a 4-week experiment every day, roughly 1 in 4 “significant” results could be false. This means you might ship features that have no real effect — or even hurt your metrics.

Key Takeaway

The peeking problem: Checking a traditional test repeatedly inflates the false positive rate far beyond its nominal level. We need a fundamentally different kind of test — one that remains valid no matter when or how often we look.

That kind of test exists. It is called a **sequential test**, and it is what Eppo (and similar platforms) use under the hood.

Act 2 — The Eppo Solution

(Schmit & Miller, 2024)

The Big Idea

Intuitive Explanation

Eppo replaces the standard confidence interval with a **sequential confidence interval** — a band around the estimated treatment effect that is valid at *every moment* since the

experiment started, not just at one pre-planned analysis time.

The key insight: a sequential CI is wider than a standard CI (you pay a small price for the freedom to peek), but the error guarantee holds no matter when you look. And Eppo reduces the noise in your data using pre-experiment information, making the CI narrower and experiments faster.

Simulation

What is being simulated: A single A/B experiment flowing through Eppo's pipeline, stage by stage.

Animated pipeline diagram. Data flows left-to-right through labelled stages. Each stage lights up as it is explained.

Randomise → Collect → Adjust → Estimate → CI → Decide

🎲 Randomise

📊 Collect

⚖️ Adjust

📈 Estimate

📏 CI

✅ Decide

EPPO Pipeline Simulation

Watch the full Eppo pipeline in action: randomise users, collect data with CUPED adjustment, build sequential confidence intervals, and see when a decision is reached.

True effect size: 0.15

-0.3 (harm) 0 (null) 0.5 (strong)

Max sample size per group: 10,000

1,000 500,000 1,000,000

Run Pipeline

The Pipeline, Step by Step

Step 1: Randomise

Each user arriving at the website is randomly assigned to **control** (existing experience) or **treatment** (new feature). Randomisation ensures the groups are comparable — any difference

in outcomes is caused by the treatment, not by pre-existing differences.

Step 2: Collect data over time

As users interact with the product, two quantities are recorded for each user:

- Y_i: the **outcome** during the experiment (e.g. purchase amount, clicks, session length).
- X_i: a **pre-experiment covariate** — the same metric from before the experiment started (e.g. purchases in the prior 2 weeks).

Step 3: Regression adjustment (noise removal)

Users are wildly different — some buy €0, others €500. This noise drowns out the treatment effect. Eppo reduces this noise by using pre-experiment data.

For each group **separately**, fit a model predicting Y from X:

$$\hat{f}_{\text{control}}(X), \qquad \hat{f}_{\text{treatment}}(X)$$

Then compute the **adjusted outcome** for each user:

$$Y_i^* = Y_i - \hat{f}_g(X_i)$$

where g is the user's group (control or treatment).

> "Predict what each user would have done based on their history, then subtract that prediction. What's left is the part you *couldn't* have predicted — and that's where the treatment effect hides."
>
> This is a generalisation of a technique called CUPED (Deng et al., 2013). The stronger the correlation ρ between pre-experiment and experiment behaviour, the more noise is removed:
>
> $$\text{Var}(Y^*) = \text{Var}(Y)(1 - \rho^2)$$
>
> At $\rho = 0.7$, half the variance is eliminated — equivalent to doubling your traffic.

Safety rule: The covariate X must come from the *pre-experiment* period. Never use data that could be affected by the treatment.

Step 4: Estimate the treatment effect

Compute the adjusted group means and the **relative lift**:

$$\hat{\tau}(t) = \hat{\mu}_1(t) - \hat{\mu}_0(t), \qquad \text{Relative lift}(t) = \frac{\hat{\tau}(t)}{\hat{\mu}_0(t)}$$

Here $\hat{\mu}_0(t)$ and $\hat{\mu}_1(t)$ are the average adjusted outcomes in the control and treatment groups at time t. Their difference $\hat{\tau}(t)$ is the estimated treatment effect. Dividing by the control mean gives a percentage: "the treatment increased purchases by +3%."

Relative lift is preferred because it is **scale-invariant** — a 3% lift means the same thing whether the baseline is €10 or €10,000.

Step 5: Estimate the variance

$$\hat{\sigma}^2_{\hat{\tau}}(t) = \frac{s_0^2(t)}{n_0(t)} + \frac{s_1^2(t)}{n_1(t)}$$

Here $s_g^2(t)$ is the sample variance (spread) of the adjusted outcomes in group g, and $n_g(t)$ is the number of users in that group so far. The variance is estimated from the data — not assumed known.

Step 6: Construct the sequential confidence interval

This is the key formula. Instead of a standard CI (which uses the constant $z_{\alpha/2} \approx 1.96$), Eppo uses a time-varying multiplier:

$$\mathrm{CI}(t) = \hat{\tau}(t) \ \pm \ \hat{\sigma}_{\hat{\tau}}(t) \cdot \underbrace{\sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}}_{\text{sequential multiplier}}$$

Unpacking the formula:

- $\hat{\tau}(t)$: the estimated treatment effect at time t (from Step 4).
- $\hat{\sigma}_{\hat{\tau}}(t)$: the estimated standard error (from Step 5).
- $n = n_0(t) + n_1(t)$: total observations so far.
- α: significance level (e.g. 0.05 for a 95% CI).
- ν: a tuning parameter set from the expected total sample size M: $\nu = M \cdot \hat{\sigma}^2$. It controls *which* sample size the CI is tightest for.
- The "sequential multiplier" replaces the constant 1.96. It starts higher (e.g. ~3.8 early on) and slowly decreases toward 1.96 — but never quite reaches it. This is the **price of peeking**.

This formula comes from the *confidence sequence* framework of Howard et al. (2021), specifically the Normal mixture boundary. The mathematical guarantee: the true treatment

effect lies inside the CI at *all times simultaneously* with probability $\geq 1 - \alpha$.

Step 7: Decide

What you see	Colour	Action
CI entirely above 0	Green	Positive effect. Ship the feature.
CI entirely below 0	Dark grey	Negative effect. Revert.
CI crosses 0, experiment ongoing	Grey	Inconclusive. Keep collecting.
CI crosses 0, experiment ended	Grey	No significant effect. Decide pragmatically.

This decision is valid at whatever time it is made. No need to pre-specify an analysis time. This solves the peeking problem from Act 1.

Key Takeaway

The Eppo pipeline in one sentence: Randomise → collect outcomes Y and covariates X → regression-adjust (remove predictable noise) → estimate lift and variance → wrap in a sequential CI → decide when CI excludes zero.

What makes it work:

- **Framework:** Howard et al. (2021) confidence sequences.
- **Noise reduction:** Generalised CUPED via per-group regression.
- **Output:** Relative lift with a CI that is valid at any time.

The Hybrid Approach

The full sequential pipeline described above is powerful but comes with a cost: the sequential CI is always wider than a fixed-horizon CI (the "price of peeking"). For the **primary KPI** — the metric you are trying to improve — this wider CI means you need more data to detect the same effect size.

Intuitive Explanation

Use sequential testing where it matters most — early stopping for safety — and fixed-horizon testing where it is most efficient — the primary metric.

The hybrid approach splits your metrics into two categories:

- **Guardrail KPIs** (revenue, error rate, latency, etc.): monitored with a **sequential CI**. You can check these every day or even continuously. If the CI goes fully below zero, abort immediately.
- **Primary KPI** (the metric you are trying to improve): analysed with a **standard fixed-horizon CI** at the pre-planned end of the experiment. No peeking penalty. Full statistical power.

What you gain

	Full sequential	Hybrid
Primary KPI CI width	Wider (~10–40%)	Standard (no penalty)
Primary KPI power	Reduced	Full
Guardrail protection	Continuous	Continuous
Early stopping for success	Yes	No
Early stopping for harm	Yes	Yes
Complexity	Higher	Lower

Key Takeaway

The hybrid approach in one sentence: Monitor guardrail KPIs with a sequential CI for early abort; analyse the primary KPI with a standard CI at the planned end-date.

This gives you the safety of sequential testing where it matters most (preventing harm) without paying the power penalty on the metric you care about most (the primary KPI).

This is the recommended approach for most A/B testing programmes.

Act 3 — What If You Don't Have Eppo?

Three progressively better approaches to peek-safe testing that anyone can implement with a spreadsheet or a few lines of code.

Intuitive Explanation

You want to check your experiment every Monday. If you use a standard 95% CI each time, you'll get false positives far too often (Act 1).

The simplest fix: **be stricter at each check.** If you plan to peek 4 times, demand stronger evidence each time. The three methods below differ in *how* they distribute that strictness across the peeks.

Think of it as a budget. You have a total error budget of $\alpha = 5\%$.

- **Bonferroni** distributes the budget equally: α/K per peek.
- **Pocock** distributes it more cleverly (using the correlation between test statistics at different peeks), but still equally across peeks.
- **O'Brien–Fleming** front-loads the budget: almost no spending early, most of it saved for the final analysis.

Eppo's approach (Act 2) doesn't need a fixed budget at all — it works for *any* number of peeks, continuously.

Choose a Method to Explore

Method 1: Bonferroni

The simplest correction. Divide α by K. One line of code.

Method 2: Pocock (1977)

Constant critical boundary. Tighter than Bonferroni by exploiting correlation.

Method 3: O'Brien–Fleming (1979)

Decreasing threshold. Nearly no penalty at the final analysis.

Head-to-Head Comparison

	Bonferroni	Pocock	OBF	Eppo
Pre-specify K?	Yes	Yes	Yes	No

	Bonferroni	Pocock	OBF	Eppo
Threshold type	Constant	Constant	Decreasing	Continuous
CI at peek 1 (K=4)	$2.50 \times \hat{\sigma}$	$2.36 \times \hat{\sigma}$	$4.05 \times \hat{\sigma}$	$\sim 3.8 \times \hat{\sigma}$
CI at final analysis (K=4)	$2.50 \times \hat{\sigma}$	$2.36 \times \hat{\sigma}$	$2.02 \times \hat{\sigma}$	$\sim 2.3 \times \hat{\sigma}$
Valid between peeks?	No	No	No	Yes
Variance reduction?	Manual	Manual	Manual	Built-in
Implementation	1 line	Lookup table	Formula	Platform

Key Takeaway

Which method should you use?

- **No platform, minimal effort:** Bonferroni. Commit to K peeks. Replace 1.96 with $z_{\alpha/(2K)}$. Done.
- **No platform, moderate effort:** O'Brien–Fleming. Nearly as efficient as Eppo at the final analysis, but requires pre-specifying K.
- **Need continuous monitoring or variance reduction:** Use a platform like Eppo.

The bottom line: even the crudest correction (Bonferroni) is *vastly* better than naive peeking. If you are checking results without any correction, you should start with Bonferroni *today*.

Implementing the Hybrid Approach Without Eppo

Act 2 described the hybrid approach: sequential CI on guardrail KPIs for early abort, standard CI on the primary KPI at the end. Here is how to implement it without a platform, using any of the three methods above.

Step 1: Classify your metrics

Category	Examples	Testing method
Primary KPI	Conversion rate, sign-ups	Fixed-horizon ($z = 1.96$)
Guardrail KPIs	Revenue, error rate, latency, crash rate	Sequential (any of the 3 methods)

Step 2: Set guardrail significance levels

If you have G guardrail KPIs, set $\alpha_{\text{guardrail}} = \alpha/G$ for each one.

Example: 3 guardrails ($G = 3$), $\alpha = 0.05$, 4 weekly peeks ($K = 4$):

$$\alpha_{\text{per guardrail}} = \frac{0.05}{3} \approx 0.0167$$

Then apply Bonferroni within each guardrail:

$$z^* = \Phi^{-1}\left(1 - \tfrac{0.0167}{2\times 4}\right) = \Phi^{-1}(0.9979) \approx 2.86$$

Step 3: Run the experiment

At each scheduled peek:

1. For **each guardrail KPI**: compute the CI using your chosen sequential method.
2. If **any** guardrail CI is entirely on the harmful side of zero: **ABORT** the experiment.
3. Otherwise: continue.

At the **end** of the experiment:

1. For the **primary KPI**: compute a standard CI with the full α:

$$\text{CI}_{\text{primary}} = \hat{\tau} \pm z_{\alpha/2} \cdot \text{SE}$$

SHIP if the CI excludes zero. Otherwise: no significant effect.

Key Takeaway

The hybrid approach without Eppo:

- Apply Bonferroni across your G guardrail KPIs: $\alpha_g = \alpha/G$.
- Within each guardrail, use O'Brien–Fleming (best) or Bonferroni (simplest).
- Test the primary KPI once at the end with a standard CI — no correction needed.
- **This is the recommended approach for teams without a sequential testing platform.**

Head-to-Head Method Comparison

Compare how Bonferroni, Pocock, O'Brien–Fleming, and Confidence Sequences behave on the same data. All methods use 12 equally-spaced peeks.

True effect: 0.20

Sample size per group: 10,000

Run Comparison

Summary

Act	Topic	Key message
1	The Peeking Problem	Checking a standard test repeatedly inflates false positives to 20–30%. You need a different kind of test.
2	The Eppo Solution	A sequential CI is valid at every moment. The **hybrid approach** uses sequential testing only for guardrail KPIs (early abort) and a standard CI for the primary KPI (full power).
3	DIY Alternatives	Bonferroni (simplest), Pocock, O'Brien–Fleming — three ways to peek safely without a platform. The hybrid approach works with any of them.

The core trade-off: if you want to peek, you must pay — your confidence interval must be wider. The question is how much wider, and the answer depends on how sophisticated a method you use:

Method	Price of peeking
No correction (naive)	Wrong answers (20–30% false positives)
Bonferroni	CI ~28–68% wider (depending on K)
Pocock	CI ~20–37% wider
O'Brien–Fleming	CI ~3% wider at final analysis
Eppo (sequential CI)	CI ~10–40% wider, valid *continuously*
Hybrid (recommended)	No penalty on primary KPI; sequential penalty on guardrails only

Mathematical Reference

Notation

Symbol	Meaning	Where used
Y_i	Outcome for user i during the experiment	Act 2, Steps 2–3
X_i	Pre-experiment covariate for user i	Act 2, Step 3 (CUPED)
Y_i^*	Adjusted outcome after regression adjustment	Act 2, Step 3
$\hat{\tau}(t)$	Estimated treatment effect at time t	Act 2, Step 4
$\hat{\mu}_0(t),\ \hat{\mu}_1(t)$	Adjusted group means at time t	Act 2, Step 4
$\hat{\sigma}_{\hat{\tau}}(t)$	Estimated standard error of treatment effect	Act 2, Step 5
$s_g^2(t)$	Sample variance in group g at time t	Act 2, Step 5
$n_0(t),\ n_1(t)$	Number of users in control / treatment at time t	Throughout
n	Total observations: n_0(t) + n_1(t)	Act 2, Step 6
α	Significance level (typically 0.05)	Throughout
ν	Tuning parameter: \nu = M \cdot \hat{\sigma}^2	Act 2, Step 6 (sequential CI)
M	Expected total sample size	Act 2, Step 6
K	Number of pre-planned analysis times (peeks)	Act 3
k	Current peek number (1 \leq k \leq K)	Act 3
t_k	Information fraction at peek k: t_k = k/K	Act 3 (OBF)
Z_k	z-statistic at peek k: \hat{\tau}/\text{SE}	Act 3
$z_{\alpha/2}$	Standard Normal critical value (\approx 1.96)	Throughout
$z_{\alpha/(2K)}$	Bonferroni-adjusted critical value	Act 3 (Bonferroni)
c_P	Pocock constant critical value	Act 3 (Pocock)
c_k^{OBF}	O'Brien–Fleming critical value at peek k	Act 3 (OBF)
Φ^{-1}	Inverse standard Normal CDF (quantile function)	Act 3

Symbol	Meaning	Where used
ρ	Correlation between pre-experiment and experiment metrics	Act 2, Step 3
G	Number of guardrail KPIs	Act 3 (Hybrid)

Key Formulas

Standard Confidence Interval

$$\text{CI} = \hat{\tau} \ \pm \ z_{\alpha/2} \cdot \text{SE}$$

Valid only at one pre-specified time. The constant $z_{\alpha/2} \approx 1.96$ for a 95% CI.

Sequential Confidence Interval (Howard et al., 2021)

$$\text{CI}(t) = \hat{\tau}(t) \ \pm \ \hat{\sigma}_{\hat{\tau}}(t) \cdot \sqrt{\frac{n+\nu}{n} \cdot \log \frac{n+\nu}{\nu\alpha^2}}$$

Valid at *all times simultaneously*. The multiplier starts high (~3.8) and decreases toward 1.96.

CUPED Variance Reduction

$$Y_i^* = Y_i - \hat{f}_g(X_i) \qquad \text{Var}(Y^*) = \text{Var}(Y)(1-\rho^2)$$

At $\rho = 0.7$, half the variance is eliminated.

Treatment Effect & Standard Error

$$\hat{\tau} = \bar{Y}_1 - \bar{Y}_0 \qquad \text{SE} = \sqrt{\frac{s_0^2}{n_0} + \frac{s_1^2}{n_1}}$$

Bonferroni Adjusted CI

$$\mathrm{CI}_k = \hat{\tau}(t_k) \pm z_{\alpha/(2K)} \cdot \mathrm{SE}(t_k)$$

O'Brien–Fleming Boundary

$$c_k^{\mathrm{OBF}} = \frac{z_{\alpha/2}}{\sqrt{t_k}} = z_{\alpha/2} \cdot \sqrt{\frac{K}{k}}$$

Pocock Boundary

$$\mathbb{P}\Big(\max_{1 \le k \le K} |Z_k| > c_P\Big) = \alpha$$

With correlation: $\mathrm{Cor}(Z_j, Z_k) = \sqrt{\min(j,k)/\max(j,k)}$

Detailed Explanations

Confidence Sequences

A confidence sequence is a sequence of confidence sets $(C_t)_{t \ge 1}$ such that the true parameter lies in C_t for *all* t simultaneously with high probability. Unlike a standard CI (valid at one time), a confidence sequence allows continuous monitoring. The price: wider intervals, especially early on.

Alpha Spending

The total false positive rate α can be thought of as a “budget” that is spent across analyses. Bonferroni spends α/K at each peek. O'Brien–Fleming spends almost nothing early and most at the end. The alpha-spending function $\alpha^*(t)$ formalises this: at information fraction t, cumulative type I error must not exceed $\alpha^*(t)$.

Information Fraction

The information fraction at peek k is $t_k = k/K$ for equally-spaced analyses, or more generally the ratio of current to planned total statistical information (proportional to sample size for equal variances). It ranges from 0 to 1 and determines where you are in the experiment.

FWER vs FDR

FWER (Family-Wise Error Rate): the probability of making *at least one* false positive across all tests. Bonferroni, Pocock, and OBF all control FWER. **FDR** (False Discovery Rate): the expected *proportion* of false positives among rejected hypotheses. FDR control is less strict and typically used when testing many hypotheses simultaneously (e.g. gene expression). For A/B testing with a small number of metrics, FWER control is standard.

References

1. **Deng, A., Xu, Y., Kohavi, R., & Walker, T.** (2013). Improving the sensitivity of online controlled experiments by utilizing pre-experiment data. *Proc. WSDM*, 123–132. — The original CUPED paper. Variance reduction via pre-experiment covariates.

2. **Howard, S.R., Ramdas, A., McAuliffe, J., & Sekhon, J.** (2021). Time-uniform, nonparametric, nonasymptotic confidence sequences. *Annals of Statistics*, 49(2), 1055–1080. — The confidence sequence framework that Eppo adopted.

3. **Pocock, S.J.** (1977). Group sequential methods in the design and analysis of clinical trials. *Biometrika*, 64(2), 191–199.

4. **O'Brien, P.C. & Fleming, T.R.** (1979). A multiple testing procedure for clinical trials. *Biometrics*, 35(3), 549–556.

5. **Schmit, M. & Miller, T.** (2024). Sequential confidence intervals for comparing two proportions with variance reduction. *Eppo Technical Report*. — Eppo's implementation combining Howard et al.'s confidence sequences with generalised CUPED.